

《数字 VLSI》第七次作业

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1. (1) 最坏情况:

$$\text{有 } t_{clk-q} + t_{logic} + \text{setup} < T - t_{skew} - 2t_{jitter}$$

$$0.4 + 1 + 0.5 < 2.5 - 0.3 - 0.1 \quad \therefore \text{满足}$$

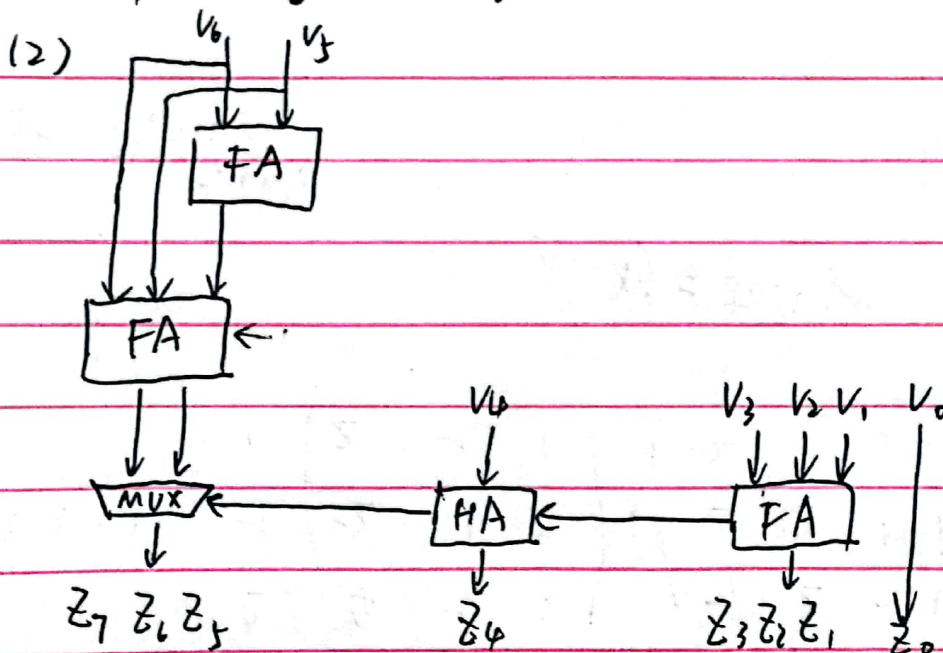
(2) 最坏情况:

$$\text{有 } t_{clk-q, cd} + t_{logic, cd} > \text{hold} + t_{skew} + 2t_{jitter}$$

$$0.2 + 0.1 > 0 + 0.3 + 0.1 \quad \therefore \text{不满足, 会出现 hold 错误}$$

2. (1) $z_0 = t, z_1 = 2t, z_2 = 3t, z_3 = 4t$

$z_4 = 5t, z_5 = 6t, z_6 = 7t, z_7 = 7t$



3. (1) $T = b + 2a(N-1) + b = 2a(N-1) + 2b$

(2) $t = b + 2aN$

(3) 若要满足 $2b + 2(N-1)a = 2Na + b$

$\Rightarrow b = 2a$